

Koshik Debanath

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RESEARCH OBJECTIVE

A highly motivated researcher with extensive experience in Natural Language Processing, Generative AI, and Deep Learning, evidenced by multiple peer-reviewed publications. Seeking to pursue a PhD to develop novel multi-modal models and explore their applications in complex reasoning and misinformation detection.

EDUCATION

Rajshahi University of Engineering & Technology

Rajshahi, Bangladesh

B.Sc. in Computer Science and Engineering CGPA: 3.27 / 4.00

Jan 2018 – Sep 2023

- **Relevant Coursework:** Linear Algebra, Data Structures and Algorithms, Object Oriented Programming, Discrete Mathematics, Database Management, Applied Statistics & Queuing Theory, Digital Image Processing, Neural Network and Fuzzy System, Artificial Intelligence, Data Mining

RESEARCH EXPERIENCE

Young Learners' Research Lab (YLRL)

Rajshahi, Bangladesh

Research Member

Jan 2024 – Present

- Conducted interdisciplinary research in NLP, Generative AI, and misinformation detection, resulting in five peer-reviewed publications.
- Collaborated on projects focusing on low-resource NLP for the Bengali language, analysis of political discourse, and classification of AI-generated text.
- Engaged in a collaborative environment to explore cutting-edge ML/DL applications, aligning with the lab's mission to foster impactful research.

SKILLS

Languages: Python (Expert), C/C++, Java, JavaScript, SQL, MATLAB

AI/ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, LangChain, Transformers, OpenCV

AI/ML Expertise: Generative AI (LLMs, RAG, Fine-tuning), NLP, Computer Vision, Deep Learning, Time Series Analysis, Prompt Engineering, Explainable AI (XAI), Data Mining

Tools & Platforms: Git, Docker, FastAPI, Flask, Django, CI/CD, MLOps, Pinecone, MongoDB, MySQL, SQLite

PUBLICATIONS

Journal Articles

- **Debanath, Koshik** and Aich, Sagor and Srizon, Azmain Yakin, “*Bayesian Physics-Informed Neural Networks for Parameter Inference and Uncertainty Quantification in Reaction-Diffusion Models of Wound Healing*,” **Under review** *Mathematical Biosciences* (July 2025). Preprint available at [SSRN](#) or [DOI](#).
- **Debanath, Koshik** and Srizon, Azmain Yakin, “*Optimizing Semantic Retrieval for Bengali: A Comparative Analysis of Monolingual and Multilingual Embeddings with Matryoshka Representation Learning*,” **Under review** *Transactions on Asian and Low-Resource Language Information Processing(TALLIP)*

Conferences

- **K. Debanath**, A. F. M. M. Rahman and M. A. Hossain, “*An Attention-Based Deep Learning Approach to Knee Injury Classification from MRI Images*,” [2023 26th International Conference on Computer and Information Technology \(ICCIT\)](#), Cox's Bazar, Bangladesh, 2023, pp. 1-6, doi: 10.1109/ICCIT60459.2023.10441340.
- **K. Debanath**, S. Aich and A. Y. Srizon, “*Advancing Low-Resource NLP: Contextual Question Answering for Bengali Language Using Llama*,” [2025 International Conference on Electrical, Computer and Communication Engineering \(ECCE\)](#), Chittagong, Bangladesh, 2025, pp. 1-6, doi: 10.1109/ECCE64574.2025.11013841.

- S. Aich, **K. Debanath** and A. Y. Srizon, “Distinguishing Between Formal and Colloquial: A Multilingual BERT Approach to Bengali Language Classification,” [2025 International Conference on Electrical, Computer and Communication Engineering \(ECCE\)](#), Chittagong, Bangladesh, 2025, pp. 1-6, doi: 10.1109/ECCE64574.2025.11013999
- **K. Debanath**, S. Aich and A. Y. Srizon, “Analyzing Bot Activity and Political Discourse in the 2024 U.S. Presidential Election: A Machine Learning Approach to Misinformation and Manipulation,” [2nd International Conference on Next-Generation Computing, IoT and Machine Learning \(NCIM-2025\)](#), Gazipur, Bangladesh, 2025, pp. 1-6, doi:10.1109/NCIM65934.2025.11160229.
- S. Aich, **K. Debanath**, and A. Y. Srizon, “Distinguishing Human-Written and AI-Generated Text: A Comprehensive Study Using Explainable Artificial Intelligence in Text Classification,” [2nd International Conference on Next-Generation Computing, IoT and Machine Learning \(NCIM-2025\)](#), Gazipur, Bangladesh, 2025, pp. 1-6, doi: 10.1109/NCIM65934.2025.11160309.
- **K. Debanath**, S. Aich, and A. Y. Srizon, “Physics-Informed Neural Networks for Real-Time Anomaly Detection in Power System Dynamics,” in Arefin, M. S., Kaiser, M. S., Bhuiyan, T., Based, M. A., and Ray, K. (eds), [Proceedings of the 3rd International Conference on Big Data, IoT and Machine Learning \(BIM 2025\)](#), Lecture Notes in Networks and Systems, vol. 1798, Springer, Cham, 2026. doi: 10.1007/978-3-032-15346-3_31.

EXPERIENCE

Manaknightdigital Inc.

Data Scientist

Toronto, ON, Canada (Remote)

Mar 2023 – Apr 2025

- **Chatbot Development:**
 - * Collected and processed product information using Excel, pandas, and openpyxl.
 - * Integrated GPT-4 to respond to user queries and manage token size limitations.
 - * Utilized libraries like nltk, sklearn, and Flask for deploying the chatbot.
- **Fraud Detection System:**
 - * Performed EDA and feature extraction on transaction datasets.
 - * Developed and optimized ML models including Xgboost, SVC, and Logistic Regression.
 - * Achieved 90% accuracy in detecting fraudulent transactions and deployed the system using Flask.
- **Data-driven ChatBot for Financial Queries:**
 - * Implemented RAG and Pinecone, enhancing data retrieval speed by 40%, enabling faster decision-making for lenders.
 - * Improved data retrieval accuracy by 25% using Cohere reranking, resulting in more precise financial advice.
 - * Applied Beautiful Soup and PyPDF2 for data scraping and processing.
- **Sports Data Analysis ChatBot:**
 - * Scraped and analyzed football data to predict match outcomes.
 - * Integrated RAG and Pinecone for efficient data querying and vector database management.
 - * Employed Beautiful Soup and PyPDF2 for data collection, analyzing 2 million football data points to achieve a 90% prediction accuracy, supporting strategic betting decisions.
- **Custom Image Generation System:**
 - * Developed an image generation platform using Stable Diffusion.
 - * Fine-tuned custom models to generate images based on user-defined presets.
 - * Utilized PyTorch and transformers for model training and deployment and finally used Docker for containerization.
- **AI-driven Data Matching System:**
 - * Organizational data was segmented using models such as Llama-2-7B and then fine-tuned to extract sections and subsections.
 - * Applied cosine similarity for matching data to specific tenders.
 - * Integrated GPT-4 for generating rationale from corresponding data.
 - * Matched organizational data against specific tenders, increasing successful tender submissions by 70%.
- **AI-Powered Collectible Authentication & Appraisal Platform:**

- * Trained deep learning models (PyTorch/TensorFlow, e.g., InceptionV3, ResNet50, CLIP) for image classification (authenticity) and similarity search.
- * Engineered an efficient CLIP+FAISS image similarity system for large-scale appraisal lookups.
- * Developed Flask/FastAPI APIs to serve model predictions (classification, similarity, appraisal).
- * Designed a multi-modal tag identification system using Serverless (RunPod API), TF-IDF, and CLIP/FAISS similarity.
- * Implemented asynchronous data pipelines (aiohttp, asyncio, pandas) for large-scale image and metadata ingestion from APIs.
- * Developed a Streamlit web application for user image uploads and displaying similarity/appraisal results via API calls.

Universal Machine Inc.

Sunnyvale, CA, USA (Remote)

Software Engineer I

Apr 2025 – Present

• YouTube Live Stream Bot:

- * Developed Chrome Extension automating YouTube Live chat using JavaScript, Chrome APIs, and async requests.
- * Integrated YouTube & OpenAI APIs for real-time chat fetching/posting and AI response generation.
- * Engineered AI features managing conversational history (chrome.storage) and prompt engineering for context/recall.
- * Implemented secure Google OAuth (chrome.identity) and robust error handling for external APIs.

• cBORG DAO Governance Platform:

- * Built a full-stack decentralized governance platform using React/Next.js, FastAPI, PostgreSQL, and Ethereum smart contracts for community proposal voting and treasury management
- * Integrated OpenAI GPT-4o to automatically parse natural language chat messages into structured trading proposals (buy/sell/hold) with confidence scoring and real-time voting
- * Implemented SIWE (Sign-In With Ethereum) wallet linking with nonce-based authentication, JWT tokens, and privacy-preserving user identity management
- * Developed live chat with proposal detection, voting dashboards, and mobile-responsive UI using Socket.io, Tailwind CSS, and modern React patterns
- * Created Solidity smart contracts for automated proposal execution and member verification, deployed on Ethereum testnet with Hardhat development framework
- * Implemented rate limiting, CORS protection, encrypted sessions with Redis, and comprehensive authentication flows for secure Web3 application deployment

PROJECTS

SynthDetect Ultra: AI Image Forensics | *Python, Streamlit, OpenCV* | [App](#)

- * Developed a comprehensive forensic analysis tool capable of distinguishing authentic photographs from AI-generated media (GANs/Diffusion) using an ensemble of physics-inspired and statistical heuristics.
- * Implemented 15+ advanced computer vision algorithms using OpenCV and SciPy, including Frequency Domain Analysis (FFT), Error Level Analysis (ELA), GLCM texture extraction, and Gradient Field anisotropy.
- * Engineered a weighted decision engine to aggregate forensic metrics into a robust confidence score, visualized via an interactive Streamlit dashboard featuring radar charts, specific artifact flags, and local SQLite history storage.
- * **Tech Stack:** Python, Streamlit, OpenCV, NumPy, Scikit-image, PyWavelets, Matplotlib, Pandas, SQLite

Obsidian AI Assistant Plugin | *TypeScript, OpenAI API* | [GitHub](#)

- * Architected and deployed a TypeScript-based Obsidian community plugin integrating OpenAI APIs for context-aware AI assistance with multi-file context, conversation history, and real-time streaming responses.
- * Designed a modular plugin architecture with clear separation of concerns, including an OpenAI service layer, secure settings management, and reusable UI components such as modal interfaces with fuzzy search.
- * Implemented production-ready UX features including async API calls, robust error handling with user notifications, keyboard shortcuts (Ctrl/Cmd+Enter), and copy or insert actions with visual feedback.
- * **Tech Stack:** TypeScript, Obsidian API, OpenAI API, esbuild, npm, async/await, DOM manipulation, event-driven architecture

AI Investment Committee for Binance | [GitHub](#) | [Streamlit App](#)

- * Designed a multi-agent AI system with specialized agents to provide cryptocurrency investment recommendations
- * **Tech Stack:** Python, OpenAI/Gemini API, Binance API, Streamlit, Pydantic

Stock Price Forecasting | [BD App](#) | [Global App](#)

- Engineered LSTM models to forecast stock prices for Bangladeshi and global markets, deployed via Streamlit.
- **Tech Stack:** Python, TensorFlow, Keras, LSTM, Pandas, Plotly, Streamlit, 'bdshare'.

AI vs Human Generated Text Detector | [WebApp](#)

- Developed an interactive web application to classify whether a given text is human-written or AI-generated
- Preprocessed and cleaned the dataset, conducted Exploratory Data Analysis (EDA), and applied feature engineering techniques
- Trained and optimized a Support Vector Classifier (SVC) using the *Machine Hack Competition* dataset: LLM Hackathon – Decoding Discourse: AI vs Human
- **Tech Stack:** Flask, scikit-learn, Python, NumPy, Pandas, Matplotlib

DataSciencePilot (RAG System) | [GitHub](#)

- Built a chat-based interface to query custom PDFs using Pinecone for vector search and LLaMA-2 for generation.
- **Tech Stack:** LangChain, Transformers, LLaMA-2, Pinecone, Python.

CVAnalyzerPro | [Streamlit App](#)

- Developed an AI tool to automatically score candidate CVs against company job requirements using LLMs.
- **Tech Stack:** OpenAI API, Gemini API, Streamlit.

UberRidePrediction | [PyPi](#) | [WebApp](#)

- Packaged an XGBoost model as a Python module to predict Uber fares and deployed it with FastAPI.
- **Tech Stack:** Scikit-learn, XGBoost, CI/CD, FastAPI, Render.

Pinecone Integration Suite | [PyPi](#)

- Authored and published two Python libraries (PineconeUtils, PineconePDFExtractor) to simplify data handling for RAG systems.
- **Tech Stack:** Pinecone, Cohere, OpenAI, PyPDF2.

Decoding AI vs Human | [WebApp](#)

- Developed a web app that allows users to determine if text was written by a human or an AI, trained on MachineHack data.
- **Tech Stack:** Scikit-learn, AWS, Render.

CaptionCraft | [StreamlitApp](#)

- Created a web application to generate image captions using the Google Gemini Pro Vision API.
- **Tech Stack:** Gemini, Streamlit, Python.

Market Price Prediction | [GitHub](#)

- Implemented and compared multiple time-series models to predict product prices.
- **Models:** ARIMA, SARIMAX, LSTM, GRU, XGBoost, Prophet.

Movie Recommendation | [WebApp](#)

- Implemented a KNN model using cosine similarity to recommend movies based on user input.
- **Tech Stack:** Scikit-learn, Pandas, Flask, Scipy.

Potato Disease Classification | [GitHub](#)

- Built a CNN model achieving near-100% accuracy in classifying potato diseases from images.
- **Tech Stack:** TensorFlow, Keras, CNN.

Diabetes Prediction | [GitHub](#)

- Constructed an Artificial Neural Network with PyTorch to predict patient diabetes status.
- **Tech Stack:** PyTorch, Flask, Unicorn, Pandas.

COMPETITIONS & ACHIEVEMENTS

Hackathon Champion at Machine Hack: Global Ranking 539 out of 8,861.

Data Science Student Championship: Secured 7th position among 1,029 participants.

LLM Hackathon (Decoding Discourse - AI vs Human): Ranked 5th out of 227 participants.

Rental Bikes Volume Prediction Hackathon: Ranked 3rd.

News Category Prediction Hackathon: Ranked 7th.

Predicting House Prices in Bengaluru: Ranked 24th out of 2,885 participants with 87% accuracy.

Subscriber Prediction Talent Search Hackathon: Ranked 26th out of 5,045 participants.

Analytics Olympiad 2022: Ranked 82nd out of 1,029 participants.

Data Science Student Championship - South Zone: Ranked 73rd out of 554 participants.

Decoding Discourse - AI vs Human: Ranked 5th out of 293 participants.

OPEN SOURCE CONTRIBUTIONS

- Contributed to **OpenLLMetry**, an open-source observability framework for LLM applications:
 - Resolved a bug where Python data classes passed as parameters were not being serialized and logged in workflows and tasks ([PR #2800](#))
 - Implemented proper serialization support for dataclasses, ensuring they are correctly captured as inputs and outputs in observability logs
 - Added automated tests to verify serialization behavior and prevent regressions
- Contributed to **OpenLLMetry** by fixing a **TypeError** in the OpenAI embeddings metrics handler caused by comparisons between **NoneType** and integers; implemented proper handling of **None** values with error logging.
 - Added automated tests to validate the fix and ensure the robustness of embeddings metrics processing
 - Improved overall stability by preventing this error from impacting workflow execution([PR #1836](#))
- Contributed to **Pinecone Canopy**, a Retrieval-Augmented Generation (RAG) framework. ([Commit](#))
- Contributed to **Khoj AI**, an open-source AI search and personal assistant platform:
 - Added copy-to-clipboard support for references in the web app reference panel
 - Implemented markdown bullet-list formatting for copied references (notes, online, and code)
 - Contribution merged into **master** as ([PR #1144](#)) with follow-up UX polish

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

[Understanding and Applying Text Embeddings](#) – *DeepLearning.AI*

[Nov 2024]

A comprehensive short course on the end-to-end development of applications using text embeddings. Key topics included:

- Fundamentals of creating, understanding, and visualizing embedding spaces.
- Leveraging embeddings for practical applications like semantic search and retrieval.
- Building a complete Q&A system (Retrieval-Augmented Generation) using Google's Vertex AI.